

# XRPL Validator — Daily Engineering Report • 2026-09-22

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- **13 commits** (xrpl-validator 12, xrpl-sdk 1)
- **26 engine findings fixed** (F357–F382)
- 1 test / vector commit
- Amendment **BatchV1\_1** held majority (since 2026-09-15 14:06 UTC); activation due ~2026-09-29 14:06 UTC
- Amendment **PermissionDelegationV1\_1** held majority (since 2026-09-21 11:18 UTC); activation due ~2026-10-05 11:18 UTC

## Engine findings fixed

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- **fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output** ( 71fd2d5 , xrpl-validator) Soak #23 receipt (#107155576 E70334ED9A24): rKVBZwTW buys 175 XLM for 37.301075 USD holding 0.958 XRP against a 1.6 XRP reserve. flowCross moved nothing the ledger could see but reported a dust output; rippled's `crossed` is `takerAmount != placeOffer` (OfferCreate.cpp:794) — the amounts were unchanged, so nothing crossed and the reserve gate claimed `tecINSUF_RESERVE_OFFER`. We set `crossed` from `actual_out > 0`, took ...
- **fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order** ( 7c9fde0 , xrpl-validator) Soak #24 receipt (#107159929 B88A96E280D3): raQWZPjc updates 9 of its oracle's 10 pairs. rippled's `OracleSet::doApply` collects every pair the oracle holds as `BaseAsset + QuoteAsset` only, then applies the transaction's entries — delete (no `AssetPrice`), update (`AssetPrice`; `Scale` only when sent), add — and emits the `std::map` order of (`BaseAsset`, `QuoteAsset`). The unnamed pair (`CLEARPOOL_AUM`) therefore loses its ...
- **fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0** ( 12a043f , xrpl-validator) `fixNFTokenPageLinks` and `fixCleanup3_2_0` are enabled on mainnet, so any mainnet `LedgerStateFix` would have applied as `tecUNSUPPORTED` against the network's code — a guaranteed receipt. Found by checking the 3.4.0 transaction list against `dispatch.rs`.
- **fix(ledger): F360 — PermissionDelegationV1\_1: DelegateSet, the delegate pays the fee, preFeeBalance\_ is the account's own** ( 0e3f28b , xrpl-validator) `PermissionDelegationV1_1` reached majority on mainnet 2026-09-21 (active ~10-05); the engine had no `DelegateSet` and no delegated fee path. rippled 3.4.0: - `Transactor::apply` consumes the `ACCOUNT`'s sequence/ticket but `payFee` charges the fee payer = `getInitiator()` = `sfDelegate` when present. `apply_common` now does the same (`TxFIELDS::delegate()`). - `preFeeBalance_` is the account's balance before the fee. Every site that ...
- **fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)** ( 28aec17 , xrpl-validator) Campaign 15 (testnet, 138 landed txs: `AccountSet` depth, `SetRegularKey`,

SignerListSet, 13 multi-signed) — 130/138 byte-exact on the live engine; the 8 misses were five rules: - F361: an unknown SetFlag/ClearFlag (32, 99) is a silent no-op; the `>= 32 → temMALFORMED` rule had no specimen behind it. - F362: the AccountTxnID prior check reads an absent field as the zero hash (getFieldH256 on a non-present optional), not ...

- **fix(ledger): F366-F369 — OracleSet time window and reserve, NFTTokenModify order, noCurrency "1" (campaign 19)** ( `0a3dfb4` , xrpl-validator) Campaign 19 (testnet, 114 landed: NFTTokenModify, NFTTokenCancelOffer, Oracle, DID depth) — 10 misses, four rules: - F366: OracleSet's  $\pm 300$  s LastUpdateTime window is centred on the header closeTime of the ledger being BUILT = parent close + closeTimeResolution, not the parent close every hasExpired test uses (3-25 tec / 3-26 tes). - F367: OracleSet tests the reserve for every adjustment (+2...-1) against ...
- **fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18)** ( `fa5735e` , xrpl-validator) Campaign 18 (testnet, 124 landed MPT txs) — 15 misses, four rules, plus one Batch-era rule found reading the same code: - F370: MPT direct payments with a TransferFee. rippled's "No rounding" means no rounding MODE: the cost is multiply(amount, rate) = Number product (16-digit mantissa, to nearest) cast to int64 (to nearest, ties even); a partial delivery is divide(SendMax, rate) = muldiv(num, 1e17, den) + 5 then to ...
- **fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)** ( `e2c806b` , xrpl-validator) Campaign 20 (testnet, 117 landed: DepositAuth by credentials across Payment, EscrowFinish, PaymentChannelClaim, CheckCash, AccountDelete) — 11 misses, six rules: - F375: the MPT payment arm runs the whole verifyDepositPreauth (expired cleanup, credential-set authorization), not only the by-account preauth. - F376: on a direct XRP payment verifyDepositPreauth — cleanup included — runs only when the amount or the ...
- **fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)** ( `081001c` , xrpl-validator) Campaign 17 (testnet, 164 landed: the OfferCreate flag matrix — Passive, IoC, FoK (fixFillOrKill), Sell, TickSize, TransferRate, self-cross, expiry, OfferSequence, unfunded/frozen makers, reserve edges, freezes) — 156/164 on the live engine; 7 misses were one rule, in two places: - checkGlobalFrozen on the TakerPays asset, then TakerGets, right after the account and ahead of funding and expiry: an issuer under ...
- **fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)** ( `9010547` , xrpl-validator) rippled's Batch::preflightSigValidated builds the required batch signers from rb.getInitiator() — the inner's sfDelegate when it carries one, because the delegate signed it — plus an sfCounterparty and a co-signing sfSponsor, each unless it is the outer account (Batch.cpp:423-445). We used the inner's Account, so a validated batch with a delegated inner would be refused here as temBAD\_SIGNER. Reachable once ...

## Engineering changes

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- **feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it** ( `c3c915c` , xrpl-validator) A zero-receipt soak only means "zero divergences" if the shadow was comparing and its compare could see a divergence. While /mnt/xrpl-data/native\_shadow.canary exists (XRPL\_NATIVE\_SHADOW\_CANARY overrides), the next ledger that wrote an AccountRoot the FFI leg agreed on gets one drop added to it in the mirror just before the real compare. The compare must flag it; the result is written as its own ...

- **fix(core): encode rippled's "1" spelling of noCurrency (finding 369)** ( c6fd55d , xrpl-sdk) rippled's to\_string(Currency) renders 0x00...01 (noCurrency) as "1"; an Oracle BaseAsset/QuoteAsset can carry it, and the shadow reads rippled JSON, so the object it writes could not be encoded (InvalidCurrency). Found by xrpl-validator campaign 19 (testnet 856F2F37D316 / 0717E2C38424).

## Tests and campaigns

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- **test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set** ( dd90d91 , xrpl-validator)

## Amendment watch

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*Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.*

| Amendment                | Majority since       | Activation due       |
|--------------------------|----------------------|----------------------|
| BatchV1_1                | 2026-09-15 14:06 UTC | 2026-09-29 14:06 UTC |
| PermissionDelegationV1_1 | 2026-09-21 11:18 UTC | 2026-10-05 11:18 UTC |

Amendments enabled on mainnet: 94.

## XRPL research

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### Automated research — candidate amendment ideas

*Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: [iceland.halcyon-names.io](https://iceland.halcyon-names.io)). They are exploration leads, not proposals.*

- **Time-Locked Account Recovery via Signer Lists** (accounts) Add an IsfRecoveryList flag settable via SignerListSet on a dedicated signer-list slot, plus two AccountRoot SFields: sfRecoveryDelay (ledgers) and sf
- **Fixed-Term Amortizing Loans With Scheduled Claims** (lending) Add sfMaturityLedger, sfInstallmentAmount, sfInstallmentInterval, and sfNextInstallmentLedger to the loan object introduced by LendingProtocolV1\_1, fi

## Appendix — commits

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| Time  | Repo           | Commit  | Subject                                                                                                                                  |
|-------|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| 07:27 | xrpl-validator | 71fd2d5 | fix(ledger): F357 — OfferCreate judges "crossed" by the after-cross amounts, not by a dust output                                        |
| 10:11 | xrpl-validator | 7c9fde0 | fix(ledger): F358 — OracleSet rebuilds the series as rippled does: unnamed pairs keep only their asset names, output in token-pair order |
| 22:19 | xrpl-validator | c3c915c | feat(node): receipt canary — the live shadow plants a 1-drop diff on demand and proves its compare still flags it                        |
| 22:43 | xrpl-validator | 12a043f | fix(ledger): F359 — LedgerStateFix was never dispatched; port both fix types from rippled 3.4.0                                          |

| Time  | Repo           | Commit  | Subject                                                                                                                                  |
|-------|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| 22:53 | xrpl-validator | 0e3f28b | fix(ledger): F360 — PermissionDelegationV1_1: DelegateSet, the delegate pays the fee, preFeeBalance_ is the account's own                |
| 23:00 | xrpl-validator | 28aec17 | fix(ledger): F361-F365 — AccountSet / SignerListSet rules from campaign 15 (first multi-signed specimens)                                |
| 23:06 | xrpl-sdk       | c6fd55d | fix(core): encode rippled's "1" spelling of noCurrency (finding 369)                                                                     |
| 23:06 | xrpl-validator | 0a3dfb4 | fix(ledger): F366-F369 — OracleSet time window and reserve, NFTokenModify order, noCurrency "1" (campaign 19)                            |
| 23:14 | xrpl-validator | dd90d91 | test(node): F360 — campaign 22 (devnet PermissionDelegation): 29 delegation vectors + DelegateSet read-set                               |
| 23:23 | xrpl-validator | fa5735e | fix(ledger): F370-F374 — MPT transfer-fee arithmetic through Number, legacy escrow divideRound, IssuanceSet/Clawback gates (campaign 18) |
| 23:30 | xrpl-validator | e2c806b | fix(ledger): F375-F380 — credential DepositAuth: MPT arm, XRP carve-out, AccountDelete, CredentialAccept/Delete order (campaign 20)      |
| 23:33 | xrpl-validator | 081001c | fix(ledger): F381 — OfferCreate preclaim's two tecFROZEN tests (campaign 17)                                                             |
| 23:49 | xrpl-validator | 9010547 | fix(ledger): F382 — a Batch's required signers come from each inner's initiator (the Delegate when present)                              |